



Efficacy and safety of cemdisiran siRNA in myasthenia gravis (NIMBLE): a double-blind, randomised, placebo-controlled, phase 3 trial

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Summary

Background Autoantibody-mediated complement activation drives pathology in acetylcholine receptor (AChR) antibody-positive generalised myasthenia gravis. Newer targeted therapies lower overall antibody concentrations or inhibit complement activity. We aimed to evaluate an siRNA (cemdisiran) targeting complement component 5 (C5) as monotherapy and in combination with a C5 antibody (pozelimab) in generalised myasthenia gravis.

Methods NIMBLE is a randomised, double-blind, placebo-controlled, phase 3 trial conducted at 86 centres in 13 countries. We enrolled patients aged 18 years or older with a diagnosis of generalised myasthenia gravis, positive serology for anti-AChR or anti-LRP4 antibodies, and a Myasthenia Gravis-Activities of Daily Living (MG-ADL) score of 6 or greater. Patients were randomly allocated to receive cemdisiran monotherapy (600 mg every 12 weeks; cemdisiran group), pozelimab monotherapy (200 mg every 4 weeks; pozelimab group), combined cemdisiran (200 mg every 4 weeks) and pozelimab (200 mg every 4 weeks; combination group), or placebo, all administered subcutaneously, during a 24-week double-blind treatment period. The primary endpoint was change from baseline in MG-ADL scores at week 24, assessed in the modified intention-to-treat (mITT) primary analysis set (the first 245 randomly allocated patients who received any dose of study treatment and had at least one post-baseline assessment). A prespecified hierarchical statistical testing strategy was used; multiplicity was controlled for comparisons of cemdisiran versus placebo and combination versus placebo (pozelimab was not statistically tested against placebo and was used only to assess the contribution of components to the combination treatment). Safety was mainly assessed by recording of treatment-emergent adverse events in all patients who received any dose of study treatment. This study is registered with ClinicalTrials.gov (NCT05070858); the current status is active, not recruiting.

Findings Between Jan 20, 2022 and July 18, 2025, 390 participants were screened and, as of the data cut-off presented in this report (July 8, 2025), 284 were randomly allocated to the cemdisiran group (79 [28%] patients), pozelimab group (50 [18%]), combination group (80 [28%]), and placebo group (75 [26%]). Of 277 patients who received any study treatment, 263 (95%) completed the double-blind treatment period. In the mITT primary analysis set, at week 24, the least-squares mean change from baseline in MG-ADL total score was -4.5 (SE 0.4) in the cemdisiran group ($n=64$), -4.0 (0.4) in the combination group ($n=67$), and -2.2 (0.5) in the placebo group ($n=59$). Placebo-adjusted least-squares mean differences in MG-ADL score at week 24 were -2.3 (SE 0.7 ; 95% CI -3.6 to -1.0 ; $p=0.0005$) in the cemdisiran group and -1.7 (0.7 ; -3.0 to -0.4 ; $p=0.0086$) in the combination group. The proportion of participants with at least one adverse event during the double-blind treatment period was 54 (69%) of 78 in the cemdisiran group, 65 (81%) of 80 in the combination group, 40 (82%) of 49 in the pozelimab group, and 54 (77%) of 70 in the placebo group. The most common adverse event in the cemdisiran group was upper respiratory tract infection (nine [12%] of 78 patients), which occurred at a similar incidence in the placebo group (eight [11%] of 70). No serious or meningococcal infections occurred in the cemdisiran group. Adverse events leading to treatment discontinuation occurred in two participants (3%) in the placebo group and one participant (2%) in the pozelimab group. No deaths occurred during the double-blind treatment period. Two deaths occurred after the double-blind treatment period, one of which was assessed as treatment-related by the investigator but not treatment-related by the sponsor.

Interpretation Cemdisiran monotherapy and combination therapy were effective in the treatment of generalised myasthenia gravis, and were generally well tolerated. Subcutaneous dosing of cemdisiran, administered every 3 months, could provide a convenient treatment approach for generalised myasthenia gravis.

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Research in context

Evidence before this study

We searched PubMed for articles published until April, 2026, using various combinations of search terms “generalized myasthenia gravis”, “C5 inhibition”, “FcRn inhibition”, “complement cascade”, “myasthenic crisis”, “Myasthenia Gravis-Activities of Daily Living”, “Quantitative Myasthenia Gravis”, and “immunosuppressive therapies”. No language or year restrictions were applied. Complement activation by the classical pathway is central to myasthenia gravis manifestations, triggered by autoantibodies to acetylcholine receptor (AChR). Uncontrolled complement activation in patients with generalised myasthenia gravis causes membrane attack complex-dependent damage and C5a-mediated inflammation at the neuromuscular junction, leading to failed neuromuscular transmission. Patients with generalised myasthenia gravis present with skeletal muscle weakness, including diplopia or ptosis, facial weakness, and swallowing difficulties. C5 inhibitors (eg, eculizumab, ravulizumab, and zilucoplan) have been tested in randomised, double-blind, placebo-controlled, phase 3 trials in patients with generalised myasthenia gravis, and approved for the treatment of generalised myasthenia gravis in several countries. Cemdisiran is a novel triantennary N-acetylgalactosamine-conjugated siRNA targeting complement C5 mRNA that reduces plasma C5 concentrations. In phase 1/2 trials, cemdisiran showed robust suppression of C5 and complement activity but, as monotherapy, did not fully control intravascular haemolysis in another complement dependent disease, paroxysmal nocturnal haemoglobinuria (PNH). However, a low-dose combination of subcutaneous cemdisiran and pozelimab (a fully human antibody targeting the C5 protein, which on its own would also not be expected to be efficacious with a low-dose regimen) in patients with PNH led to a more complete complement blockade that improved efficacy compared with a standard-of-care C5 inhibitor, ravulizumab, from an initial exploratory cohort in an ongoing phase 3 study. In generalised myasthenia

gravis, the concept that complete inhibition of the complement pathway is necessary to achieve optimal clinical response in patients is supported by complement activity data of a peptide inhibitor of C5, zilucoplan, from a phase 2 clinical trial; however, it has not been fully validated. While targeted therapies are available, there remains an unmet need for treatments that provide improved efficacy across dosing intervals, with a favourable safety profile and reduced treatment burden.

Added value of this study

The phase 3 NIMBLE study evaluated the efficacy and safety of cemdisiran monotherapy administered every 3 months by subcutaneous injection, and a lower dose of cemdisiran in combination with low-dose pozelimab administered subcutaneously every month. The study included patients with AChR autoantibody-positive and LRP4 autoantibody-positive generalised myasthenia gravis, who generally require advanced therapies. The study design allowed measurement of the contribution of each component (cemdisiran and pozelimab) in the treatment of generalised myasthenia gravis. Cemdisiran monotherapy showed robust efficacy with partial inhibition of complement activity, with no meningococcal infections and no evidence of increased infective risk compared with placebo.

Implications of all the available evidence

This study showed that cemdisiran monotherapy had rapid, sustained, and clinically meaningful effects compared with placebo for the treatment of generalised myasthenia gravis, with no waning of efficacy near the end of respective 3-month dosing intervals. Cemdisiran monotherapy could provide a new approach targeting the complement C5 pathway, specifically C5 mRNA in hepatocytes. Although this regimen only led to partial complement inhibition, it could reduce treatment burden, with convenient subcutaneous dosing every 3 months. The underlying mechanisms explaining the residual complement activity while improving clinical outcomes in generalised myasthenia gravis warrant future investigations.

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See Online for appendix 1

Introduction

Generalised myasthenia gravis is a rare, chronic, autoantibody-mediated disease affecting the neuromuscular junction.^{1,2} Patients present with diplopia and/or ptosis, facial weakness, dysphagia, and skeletal muscle weakness.² Life-threatening myasthenic crisis occurs in around 15–20% of patients, mostly triggered by respiratory infection and often managed by mechanical ventilation.³

In acetylcholine receptor (AChR) antibody-positive generalised myasthenia gravis, pathology at the neuromuscular junction is driven by activation of the classical complement pathway, leading to the endplate damage.⁴ Emerging biological therapies for generalised myasthenia gravis include those that generally lower overall antibody concentrations (eg, therapies targeting neonatal Fc receptors [FcRn] or CD19⁺ B cells),^{5,6} or those

that lower complement activity (eg, therapies targeting complement component 5 [C5]).^{7–11} These therapies both appear to be immunosuppressive: FcRn antibodies (eg, nivalimab) administered chronically can activate latent Epstein–Barr virus in rare instances,¹² and C5 antibodies are associated with a risk of serious meningococcal infections.^{13–15} Furthermore, due to the high plasma concentrations of C5, antibodies blocking this target require very high-dose intravenous administrations or frequent subcutaneous injections. Thus, there remains an unmet need for targeted therapies providing rapid, sustained, and improved efficacy without waning of effect throughout dosing intervals, with a favourable safety profile and a reduced treatment burden.

Cemdisiran is an investigational, subcutaneously administered small interfering RNA (siRNA) targeting

C5 mRNA, reducing plasma C5 concentrations without binding or directly inhibiting C5. It is covalently linked to a triantennary N-acetylgalactosamine (GalNAc) ligand, targeting cemdisiran to hepatocytes (thereby reducing C5 production; appendix 1 p 24). GalNAc-siRNA conjugates generally have a more favourable safety profile (in healthy humans and those with other complement-mediated diseases)^{16,17} and have a longer duration of effect than previously used lipid nanoparticle encapsulated siRNAs, while also offering convenient subcutaneous dosing.¹⁸ Pozelimab is an investigational fully human monoclonal antibody^{19,20} that binds and blocks C5, which is approved to treat CD55-deficient protein-losing enteropathy.^{21,22}

Cemdisiran monotherapy (which alone lowers C5 concentrations by >90%) was not efficacious in paroxysmal nocturnal haemoglobinuria (PNH), another C5-dependent disease.^{16,23} However, a combination of low-dose subcutaneous cemdisiran and low-dose subcutaneous pozelimab (which on its own would also not be expected to be efficacious), given in a monthly dosing regimen to patients with PNH, allowed for deeper complement blockade that improved efficacy compared with a standard-of-care C5 antibody from an initial exploratory cohort in an ongoing pivotal phase 3 study.²⁴ We therefore reasoned that combining subcutaneous cemdisiran and pozelimab might also allow for deeper complement blockade that could improve efficacy in generalised myasthenia gravis and also allow for a convenient monthly dosing regimen.

We aimed to evaluate the efficacy of a combination of cemdisiran and pozelimab, as well as cemdisiran monotherapy, allowing us to measure the contribution of each component in the treatment of generalised myasthenia gravis.

Methods

Study design

NIMBLE is a phase 3, multinational, randomised, double-blind, double-dummy, placebo-controlled trial in participants with symptomatic generalised myasthenia gravis, conducted at 86 clinical research centres and hospitals in 13 countries (appendix 1 pp 3–14). The trial consists of a 24-week placebo-controlled double-blind treatment period, a 28-week double-blind extension treatment period, a 68-week open-label treatment period, and a 52-week post-treatment follow-up period (appendix 1 p 25). This report presents the prespecified primary efficacy and safety findings from the double-blind treatment period. The trial protocol is available in appendix 2 and statistical analysis plan in appendix 3.

The trial was conducted in accordance with the International Council for Harmonisation E6 Guideline for Good Clinical Practice, the principles of the Declaration of Helsinki, and all applicable regulatory requirements. The trial protocol and its amendments (appendix 2 pp 3–17) were approved by an institutional

review board and/or ethics committee at each participating site (appendix 1 pp 3–14). All participants provided written informed consent. Participant safety was monitored by an independent data monitoring committee.

This study is registered with ClinicalTrials.gov (NCT05070858).

Participants

Eligible participants were those aged 18 years or older with a documented diagnosis of generalised myasthenia gravis and positive serology for anti-AChR or anti-lipoprotein receptor-related protein 4 (LRP4) antibodies (these participants might benefit from complement inhibition, as it is likely that complement is involved in the pathogenesis of patients with generalised myasthenia gravis who have anti-LRP4 antibodies), symptoms of generalised myasthenia gravis as defined by a Myasthenia Gravis-Activities of Daily Living (MG-ADL) score of 6 or greater and Myasthenia Gravis Foundation of America (MGFA) clinical class II to IVa, and concurrent treatment with a stable-dose regimen of an acetylcholinesterase inhibitor and immunosuppressive therapy for myasthenia gravis (or a documented reason for not receiving therapy). The first dose of a meningococcal vaccination (serotype ACWY and serotype B [if available]) was required before randomisation. Sex (male or female), race (American Indian or Alaskan Native; Asian; Black or African American; Native Hawaiian or other Pacific Islander; White; or other), and ethnicity (Hispanic or Latino; or not Hispanic or Latino) were self-reported by the participant. Full inclusion and exclusion criteria are provided in the protocol (appendix 2 pp 77–81).

Randomisation and masking

Patients were initially randomly allocated in a 1:1:1:1 ratio to receive pozelimab monotherapy (pozelimab group), cemdisiran monotherapy (cemdisiran group), pozelimab and cemdisiran (combination group), or placebo (placebo group). When 50 participants had been allocated to the pozelimab group, further enrolment was limited to the other three groups (1:1:1 randomisation). When the placebo group reached approximately 65 participants, randomisation proceeded 2:2:1 to the cemdisiran, combination, and placebo groups. Randomisation was stratified by presence of MGFA clinical class II at screening (yes or no) and by treatment with immunosuppressive therapies (none or discontinued; all immunosuppressive therapies <1 year duration; or any immunosuppressive therapy ≥1 year duration) on day 1. Block randomisation was used to assign participants to each treatment via Interactive Response Technology.

All participants, the principal investigators, and study site personnel were masked to group allocation; masking was maintained with use of dummy drugs (pozelimab placebo and cemdisiran placebo).

See Online for appendices 2 and 3

Procedures

All study treatments were administered subcutaneously by study site personnel. Throughout the 24-week double-blind treatment period, patients in the pozelimab group received pozelimab 200 mg and cemisiran placebo at baseline (day 1) and every 4 weeks; those in the cemisiran group received cemisiran 600 mg at baseline and weeks 12, and 24, cemisiran placebo at weeks 4, 8, 16, and 20, and pozelimab placebo at baseline and every 4 weeks; those in the combination group received pozelimab 200 mg and cemisiran 200 mg at baseline and every 4 weeks; and those in the placebo group received pozelimab placebo and cemisiran placebo at baseline and every 4 weeks.

Rescue therapy was allowed in the event of myasthenic crisis or impending crisis.

Efficacy during the double-blind treatment period was assessed with validated outcome measures: MG-ADL,^{25,26} Quantitative Myasthenia Gravis (QMG),²⁷ Myasthenia Gravis Composite (MGC),²⁸ and Myasthenia Gravis Quality of Life 15-item (revised; MG-QOL15r).^{29,30} The MG-ADL is a clinician-administered, patient-reported outcome measure assessing daily functions impacted by generalised myasthenia gravis; total score ranges from 0 to 24 with higher scores indicating greater functional impairment and disability. The QMG is a 13-item performance test that assesses a person's current signs and symptoms severity of myasthenia gravis; QMG total scores range from 0 to 39 with higher scores representing greater impairment. The MGC scale consists of ten items that measure symptoms and signs of myasthenia gravis; scores range from 0 to 50, with higher scores indicating higher impairment. The MG-QOL15r is a self-administered, validated patient-reported outcome instrument consisting of 15 items that measure a person's quality of life as related to myasthenia gravis; total score ranges from 0 to 30, with higher scores representing greater impairment. Additional details of these measures are described in appendix 1 (p 16). These efficacy assessments were collected at baseline and at weeks 2, 4, 8, 12, 16, 20, and 24. The MG-ADL, QMG, and MGC were administered by a trained evaluator; the MG-QOL15r was completed by the participant.

Details of pharmacokinetic, pharmacodynamic, and immunogenicity analysis methods are described in appendix 1 (p 16–17).

Outcomes

The primary efficacy endpoint was change from baseline in MG-ADL total score at week 24. The key secondary efficacy endpoint was change from baseline in QMG total score at week 24.

Other prespecified secondary efficacy endpoints included MG-ADL response (defined as a ≥ 3 -point reduction in MG-ADL total score from baseline at week 24),⁹ QMG response (defined as a ≥ 5 -point reduction in QMG total score from baseline at week 24),⁹

change from baseline in MGC total score at week 24, change from baseline in MG-QOL15r total score at week 24, consistent response on the MG-ADL (defined as a ≥ 2 -point reduction at two or more consecutive assessments spanning ≥ 4 weeks during the double-blind treatment period, based on previously reported thresholds of clinically meaningful improvement),³¹ and minimal symptom expression (defined as a score of 0 or 1 on the MG-ADL, based on previously published assessments of therapeutic effectiveness in people with myasthenia gravis)^{11,32} at week 24.

We also assessed the following prespecified exploratory efficacy endpoints: number of myasthenic crises (including impending crises); number of any hospitalisations; total number of days of hospitalisation; and use of rescue therapy.

Safety assessments included treatment-emergent adverse events (graded on a scale of mild, moderate, or severe), clinical laboratory tests, electrocardiograms, physical examination, bodyweight, vital signs, and suicidal ideation and behaviour. Immunogenicity was measured by the incidence of antidrug antibodies to cemisiran and pozelimab over time.

Pharmacokinetic outcomes included concentrations of cemisiran and its predominant active metabolite in plasma and concentration of pozelimab in serum. Pharmacodynamic outcomes included percentage change from baseline in total complement haemolysis activity (measured as CH50)³³ in serum and concentrations of total C5 in plasma. A full list of outcomes is provided in the protocol (appendix 2 pp 60–61).

Statistical analysis

The prespecified hierarchical statistical analysis plan (appendix 3) provided for multiplicity-corrected comparisons of combination versus placebo, and cemisiran monotherapy versus placebo; the formal analysis plan did not evaluate the pozelimab versus placebo, which was included solely to confirm the contribution of components (the low pozelimab dose used was expected to minimally inhibit C5 as a monotherapy, and to only show efficacy in the setting of the lower C5 concentrations in combination with cemisiran).

Power calculations based on the primary endpoint indicated that 65 participants per group (placebo, combination, and cemisiran groups) from concurrently randomised participants were required to achieve 88% power to detect a treatment difference of -2.3 between combination and placebo at the overall type I error rate of $\alpha=0.05$, assuming an SD of 4 and an attrition rate of 7%.⁹ Using the same SD and attrition rate, the power was 81% to detect a similar treatment difference between cemisiran and placebo at $\alpha=0.025$.

We defined a modified intention-to-treat population as all randomly allocated participants who received any amount of study drug and had at least one post-baseline

efficacy assessment in the double-blind treatment period. Efficacy analyses of primary and all secondary endpoints included in the hierarchical testing procedure (appendix 1 p 26) were conducted in the mITT primary analysis set, defined as all participants who met mITT criteria among the first 245 participants to be randomly allocated in the double-blind treatment period.

The same analysis prespecified for the primary endpoint was used to assess the contribution of each monotherapy to the combination therapy. The efficacy of cemdisiran monotherapy versus combination therapy (ie, the contribution of pozelimab to combination therapy) was assessed using the mITT analysis set. The efficacy of pozelimab monotherapy versus combination

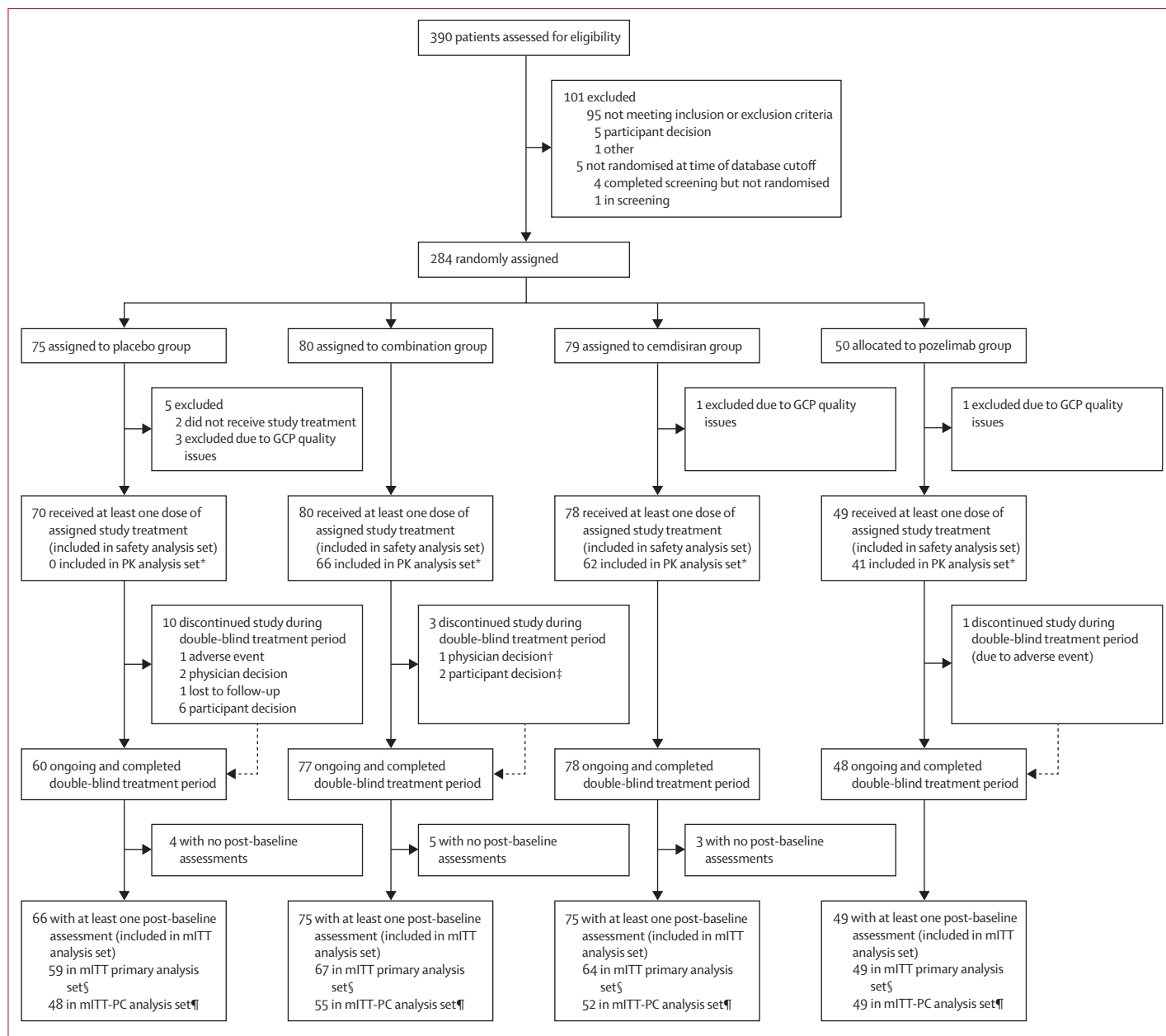


Figure 1: Flow diagram for the double-blind treatment period

GCP=Good Clinical Practice. mITT=modified intention-to-treat. mITT-PC=mITT pozelimab-concurrent. PK=pharmacokinetic. *The PK analysis set consisted of all patients who received at least one dose of active study drug and had at least one drug concentration result after receiving the first dose. †One participant was discontinued by physician decision in conjunction with a psychiatric consultation of the participant's suicidal ideation (expressed while hospitalised for myasthenic crisis). ‡Participants withdrew consent from the study (one due to an unspecified personal or family situation, and one due to feeling worse). §mITT primary analysis set consisted of those who received at least one dose of treatment and had at least one post-baseline assessment among the first 245 patients to be randomly allocated. ¶mITT-PC analysis set consisted of all patients who were randomly allocated on or before the date on which the 50th patient was allocated to the pozelimab group, received at least one dose of treatment, and had at least one post-baseline assessment.

therapy (ie, the contribution of cemisiran to combination therapy) was assessed using the mITT pozelimab-concurrent (mITT-PC) analysis set, which included all participants within the mITT who were randomly allocated on or before the date on which the 50th participant was randomly allocated in the pozelimab monotherapy group.

The assessment of the contribution of components was conducted through 95% CIs comparing the combination therapy to each of the monotherapies. For cemisiran monotherapy versus combination therapy, assuming an SD of 4.0 and a dropout rate of around 7%, 50 participants per group would yield a 95% CI halfwidth of 1.65 points. For pozelimab monotherapy versus combination therapy, assuming an SD of 4.0 and a dropout rate of around 7%, 50 participants per group would yield a 95% CI halfwidth of 1.65 points.

The efficacy of primary and continuous secondary endpoints was assessed by a linear mixed model for repeated measures (MMRM) with treatment, visit, baseline total score, randomisation stratification factors (MGFA clinical class II at screening and treatment with immunosuppressive therapy on day 1), treatment-by-visit, and baseline total score-by-visit as fixed effects; a prespecified stratification factor, region at screening (Japan or rest of the world), was removed from the MMRM due to the small number of participants in Japan. Values after intercurrent events, including study discontinuation due to death, lack of efficacy, adverse event, and use of rescue therapies, were imputed by the worst observation carried forward plus random noise. The summary of intercurrent events during the double-blind treatment period is provided in appendix 1 (p 20). Imputation was done 50 times with results summarised using the Rubin formula.³⁴ The same MMRM model was used to assess the contribution of each monotherapy to the combination. The efficacy analyses for all binary secondary endpoints, pharmacokinetic analyses, and key exploratory endpoints are described in appendix 1 (pp 16–18). Pharmacokinetic analyses were performed in the pharmacokinetic analysis set, which was defined for each study drug separately and included all participants who received any amount of active study treatment (pozelimab, cemisiran, or both) and who had at least one non-missing drug concentration result following the first dose of the respective study treatment.

All safety analyses were done in the safety analysis set, which included all randomly allocated participants who received any amount of study drug.

The cutoff date for data presented in this report was July 8, 2025. Analyses were conducted using SAS version 9.4.

Role of the funding source

The funder of the study (Regeneron Pharmaceuticals) was involved in the study design, data collection, data analysis, data interpretation, writing of the report, and

	Placebo (n=59)	Combination (n=67)	Cemisiran (n=64)	Pozelimab (n=49)
Age, years				
Mean	49.5 (15.6)	48.7 (15.1)	52.7 (15.7)	51.0 (15.6)
<35	12 (20%)	13 (19%)	10 (16%)	11 (22%)
35–64	33 (56%)	45 (67%)	34 (53%)	27 (55%)
≥65	14 (24%)	9 (13%)	20 (31%)	11 (22%)
Region				
Japan	3 (5%)	2 (3%)	3 (5%)	2 (4%)
Rest of world	56 (95%)	65 (97%)	61 (95%)	47 (96%)
Ethnicity				
Hispanic or Latino	4 (7%)	6 (9%)	8 (13%)	3 (6%)
Not Hispanic or Latino	53 (90%)	56 (84%)	54 (84%)	44 (90%)
Not reported	0	2 (3%)	0	0
Missing*	2 (3%)	3 (4%)	2 (3%)	2 (4%)
Race				
White	31 (53%)	40 (60%)	35 (55%)	29 (59%)
Black or African American	1 (2%)	5 (7%)	4 (6%)	0
Asian	24 (41%)	16 (24%)	22 (34%)	17 (35%)
American Indian or Alaska Native	0	0	0	1 (2%)
Multiple	1 (2%)	2 (3%)	1 (2%)	0
Not reported	2 (3%)	4 (6%)	2 (3%)	2 (4%)
Sex				
Female	35 (59%)	42 (63%)	36 (56%)	24 (49%)
Male	24 (41%)	25 (37%)	28 (44%)	25 (51%)
Height, cm	166.25 (10.49)	167.00 (11.47)	168.02 (9.70)	167.65 (10.69)
Weight, kg	80.11 (21.61)	80.41 (23.29)	80.86 (23.37)	82.01 (26.05)
Duration of myasthenia gravis since diagnosis, years	8.16 (9.21)	6.10 (7.14)	8.10 (8.69)	8.50 (7.56)
Number of previous myasthenic crises or impending crises				
0	39 (66%)	47 (70%)	39 (61%)	30 (61%)
1	12 (20%)	13 (19%)	14 (22%)	14 (29%)
2	5 (8%)	4 (6%)	7 (11%)	3 (6%)
3	1 (2%)	3 (4%)	2 (3%)	2 (4%)
>3	2 (3%)	0	2 (3%)	0
Time since most recent crisis, months	22.94 (23.14)	23.45 (30.92)	30.27 (51.28%)	49.94 (50.04)
Myasthenia gravis-related hospitalisation in the 2 years before randomisation				
Yes	22 (37%)	22 (33%)	24 (38%)	19 (39%)
No	37 (63%)	45 (67%)	40 (63%)	30 (61%)
Mean	1.4 (1.0)	1.8 (1.1)	1.5 (0.7)	1.3 (0.7)
History of positive serological test for anti-AChR antibodies†				
Yes	55 (93%)	63 (94%)	60 (94%)	48 (98%)
No	4 (7%)	4 (6%)	4 (6%)	1 (2%)
History of positive serological test for anti-LRP4 antibodies†				
Yes	8 (14%)	6 (9%)	5 (8%)	7 (14%)
No	51 (86%)	61 (91%)	59 (92%)	42 (86%)
History of serological test for anti-AChR antibodies and anti-LRP4 antibodies‡				
Both negative	1 (2%)	0	1 (2%)	0
Both positive	5 (8%)	2 (3%)	2 (3%)	6 (12%)
Only positive for anti-AChR antibodies	50 (85%)	61 (91%)	58 (91%)	42 (86%)
Only positive for anti-LRP4 antibodies	3 (5%)	4 (6%)	3 (5%)	1 (2%)

(Table 1 continues on next page)

	Placebo (n=59)	Combination (n=67)	Cemdisiran (n=64)	Pozelimab (n=49)
(Continued from previous page)				
MGFA classification§				
Ila	12 (20%)	18 (27%)	12 (19%)	12 (24%)
Ilb	12 (20%)	10 (15%)	14 (22%)	5 (10%)
IIla	18 (31%)	23 (34%)	23 (36%)	16 (33%)
IIlb	14 (24%)	15 (22%)	13 (20%)	13 (27%)
IVa	3 (5%)	1 (1%)	2 (3%)	3 (6%)
Number of immunosuppressive therapies used				
0	16 (27%)	14 (21%)	14 (22%)	10 (20%)
1	18 (31%)	31 (46%)	25 (39%)	22 (45%)
2	25 (42%)	21 (31%)	25 (39%)	17 (35%)
≥3	0	1 (1%)	0	0
History of complement inhibitor therapies				
Yes	4 (7%)	1 (1%)	4 (6%)	3 (6%)
No	55 (93%)	66 (99%)	60 (94%)	46 (94%)
History of FcRn inhibitor therapy				
Yes	3 (5%)	4 (6%)	3 (5%)	6 (12%)
No	56 (95%)	63 (94%)	61 (95%)	43 (88%)
MG-ADL total score	9.4 (2.6)	9.4 (2.9)	8.9 (2.4)	9.8 (3.4)
QMG total score	15.5 (6.2)	15.6 (5.5)	15.5 (4.9)	16.1 (6.1)
MGC total score	17.5 (7.2)	17.3 (6.6)	17.2 (5.8)	18.4 (8.3)
MG-QOL15r total score	16.4 (6.1)	16.3 (5.6)	15.9 (5.8)	16.0 (6.7)
Data are mean (SD) or n (%). AChR=acetylcholine receptor. FcRn=neonatal Fc receptor. LRP4=lipoprotein receptor-related protein 4. MG-ADL=Myasthenia Gravis-Activities of Daily Living. MG-QOL15r=Myasthenia Gravis Quality of Life 15-item (revised). MGC=Myasthenia Gravis Composite. MGFA=Myasthenia Gravis Foundation of America. QMG=Quantitative Myasthenia Gravis. *Some countries did not report ethnicity. †Refers to both documented historical positivity before screening and positive result in central laboratory during screening. ‡Generated post-hoc. §MGFA classification refers to baseline classification at screening; participants were not reclassified after randomisation.				
Table 1: Demographics and baseline characteristics (modified intention-to-treat primary analysis set)				

decision to submit the paper for publication. The trial was conducted in collaboration with the sponsor and the NIMBLE trial investigators.

Results

Patients were recruited into the study between Jan 20, 2022 and July 18, 2025. 390 participants were screened, of whom 101 were excluded, five had not been randomly allocated at the time of database cutoff, and 284 were randomly allocated: 75 (26%) to the placebo group, 80 (28%) to the combination group, 79 (28%) to the cemdisiran group, and 50 (18%) to the pozelimab group (figure 1). Of those randomly allocated, 277 (98%) participants received at least one dose of assigned study treatment and comprised the safety population; 265 (93%) also had at least one post-baseline assessment within the double-blind treatment period and comprised the mITT population; and 239 (84%) were included in the mITT primary analysis set. Among the 277 patients who received study treatment, 14 (5%) discontinued the study before the end of the double-blind treatment period and 263 (95%) completed this treatment period (figure 1).

Baseline demographics and disease characteristics were generally balanced across treatment groups (table 1 [mITT primary analysis set], appendix 1 pp 21–22 [mITT analysis set]). The mITT primary analysis set consisted of 137 (57%) women and 102 (43%) men, with a mean age of 50.5 years (SD 15.5). 270 (98%) participants received at least one dose of quadrivalent meningococcal vaccine within 5 years before randomisation and 160 (58%) participants received serotype B meningococcal vaccine within 3 years before randomisation.

At week 24, the least-squares mean change from baseline in MG-ADL total score (the primary endpoint) was -4.5 (SE 0.4) in the cemdisiran group, -4.0 (0.4) in the combination group, and -2.2 (0.5) in the placebo group among patients in the mITT primary analysis set (table 2). Placebo-adjusted differences in least-squares means were -2.3 (SE 0.7; 95% CI -3.6 to -1.0 ; $p=0.0005$) in the cemdisiran group and -1.7 (0.7; -3.0 to -0.4 ; $p=0.0086$) in the combination group. Clinically meaningful improvements in MG-ADL total score (defined as ≥ 2 -point reduction) occurred within 2 weeks in the cemdisiran and combination groups, deepened over time, and were sustained throughout the double-blind treatment period (figure 2A). Per the statistical analysis plan, pozelimab monotherapy was not statistically tested against placebo, but showed little effect at the studied dose (table 2). Pozelimab results are presented in the figures and tables but are not discussed in detail since pozelimab monotherapy was included solely to confirm the contribution of components.

Least-squares mean change from baseline at week 24 in QMG total score (the key secondary endpoint) was -4.2 (SE 0.6) in the cemdisiran group, -3.3 (0.6) in the combination group, and -1.5 (0.7) in the placebo group (table 2). Placebo-adjusted differences in least-squares means were -2.8 (SE 0.9; 95% CI -4.5 to -1.1 ; $p=0.0015$) in the cemdisiran group and -1.9 (0.9; -3.6 to -0.1 ; $p=0.035$) in the combination group. Reduction in QMG total scores also occurred within 2 weeks in both treatment groups, deepened over time, and were sustained throughout the double-blind treatment period (figure 2B).

Additional secondary endpoints are shown in table 2 and figure 2. The proportions of participants with responses on the MG-ADL and QMG in the cemdisiran and combination groups were significantly greater than those in the placebo group. Change from baseline at week 24 in MGC total score did not meet statistical significance for the comparison of combination versus placebo; p values are therefore not reported for the subsequent endpoints in the statistical testing strategy, including MG-QOL15r (figure 2; table 2).

Among the exploratory endpoints, the number of participants with myasthenic crisis and the number administered rescue therapy were lower in the cemdisiran group than in the combination and placebo groups (table 3). The cemdisiran group also had the lowest proportion of participants hospitalised for any

	Placebo (n=59)	Combination (n=67)	Cemdisiran (n=64)	Pozelimab (n=49)*
Primary endpoint				
Change from baseline at week 24 in MG-ADL total score				
Mean (SD)	-2.6 (3.0)	-4.3 (3.5)	-4.3 (3.3)	-2.9 (4.0)
LS mean (SE)	-2.2 (0.5)	-4.0 (0.4)	-4.5 (0.4)	-2.8 (0.5)
Placebo-adjusted difference				
LS mean (95% CI)	..	-1.7 (-3.0 to -0.4)	-2.3 (-3.6 to -1.0)	NA
p value	..	0.0086	0.0005	NA
Key secondary endpoint				
Change from baseline at week 24 in QMG total score				
Mean (SD)	-2.0 (4.6)	-3.6 (4.3)	-4.1 (4.8)	-1.9 (4.2)
LS mean (SE)	-1.5 (0.7)	-3.3 (0.6)	-4.2 (0.6)	-1.5 (0.7)
Placebo-adjusted difference				
LS mean (95% CI)	..	-1.9 (-3.6 to -0.1)	-2.8 (-4.5 to -1.1)	NA
p value	..	0.035	0.0015	NA
Other secondary endpoints				
Proportion with MG-ADL response†				
Patients, n (%)	26 (44%)	44 (66%)	49 (77%)	24 (49%)
RR (95% CI)	..	1.514 (1.071 to 2.138)	1.843 (1.331 to 2.552)	NA
p value	..	0.014	0.0001	NA
Proportion with QMG response‡				
Patients, n (%)	11 (19%)	24 (36%)	31 (48%)	9 (18%)
RR (95% CI)	..	1.929 (1.017 to 3.657)	2.678 (1.447 to 4.958)	NA
p value	..	0.037	0.0006	NA
Change from baseline to week 24 in MGC total score				
Mean (SD)	-4.6 (7.9)	-6.4 (6.6)	-7.4 (6.4)	-3.9 (7.6)
LS mean (SE)	-3.4 (1.0)	-6.0 (0.9)	-7.8 (0.9)	-3.2 (1.0)
Placebo-adjusted difference				
LS mean (95% CI)	..	-2.6 (-5.2 to 0.0)	-4.3 (-6.9 to -1.8)	NA
p value	..	0.048§	NA¶	NA
Change from baseline to week 24 in MG-QOL15r total score				
Mean (SD)	-2.9 (5.4)	-4.3 (6.2)	-4.5 (6.6)	-1.6 (5.8)
LS mean (SE)	-2.3 (0.9)	-4.3 (0.8)	-4.7 (0.8)	-1.6 (0.9)
Placebo-adjusted difference				
LS mean (95% CI)	..	-2.0 (-4.2 to 0.3)	-2.4 (-4.7 to -0.2)	NA
p value	..	NA¶	NA¶	NA
Proportion with minimal symptom expression on MG-ADL				
Patients, n (%)	4 (7%)	15 (22%)	13 (20%)	7 (14%)
RR (95% CI)	..	3.676 (1.194 to 11.318)	3.390 (1.047 to 10.969)	NA
p value	..	NA¶	NA¶	NA
Proportion with consistent response on MG-ADL**				
Patients, n (%)	37 (63%)	55 (82%)	52 (81%)	31 (63%)
Estimated OR (95% CI) vs placebo	..	1.309 (1.041 to 1.647)	1.308 (1.041 to 1.644)	NA
p value	..	NA¶	NA¶	NA

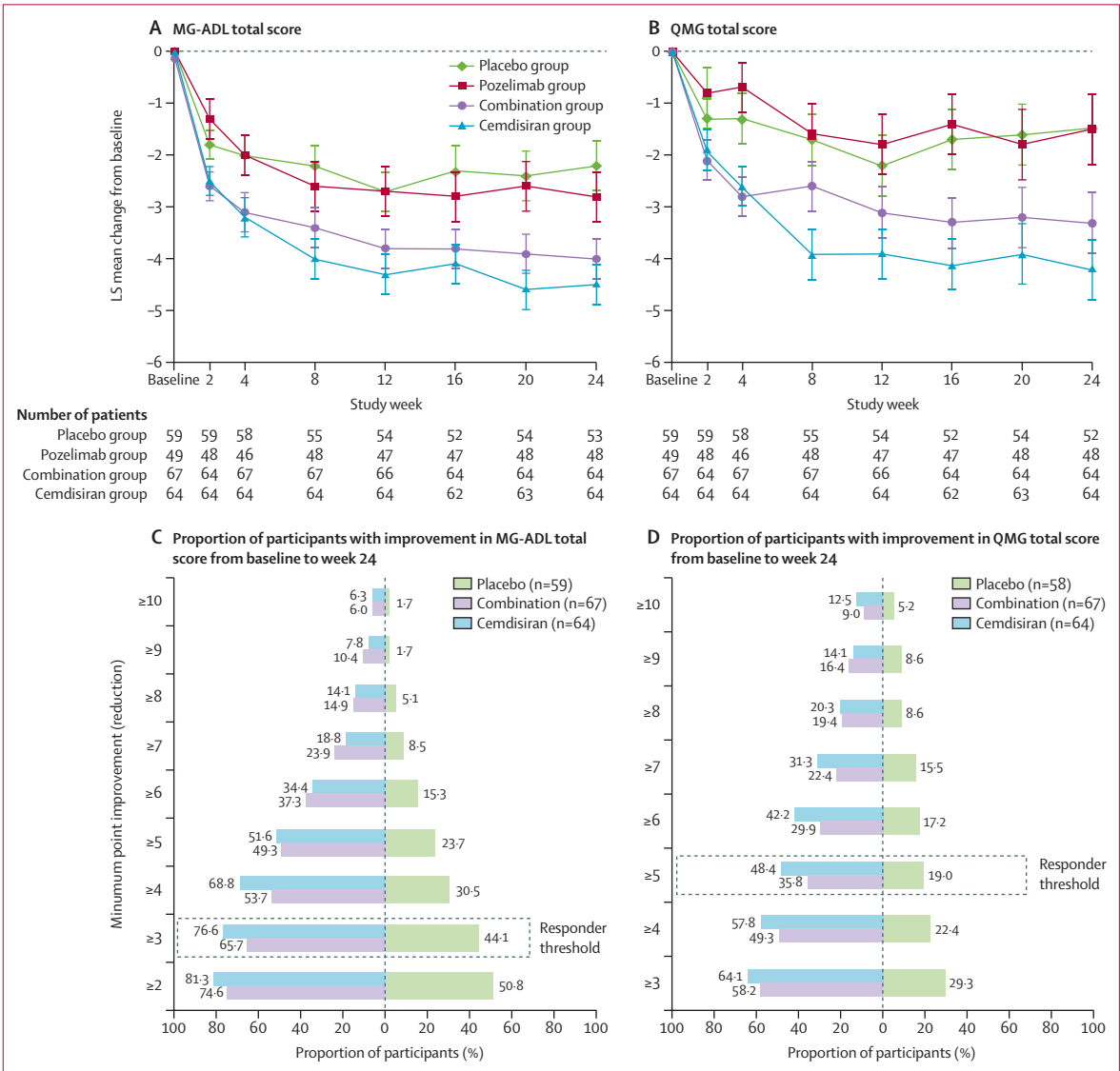
Endpoints are presented in order of the prespecified statistical testing strategy. 95% CI widths were not adjusted for multiplicity and should not be used in place of hypothesis testing. 95% CIs were derived from the linear mixed model for repeated measures model. LS=least-squares. MG-ADL=Myasthenia Gravis—Activities of Daily Living. MG-QOL15r=Myasthenia Gravis Quality of Life 15-item (revised). MGC=Myasthenia Gravis Composite. NA=not applicable. OR=odds ratio. QMG=Quantitative Myasthenia Gravis. RR=relative risk. *Placebo-adjusted differences, RRs, and corresponding p values for pozelimab monotherapy versus placebo are not presented as this comparison was not in the scope of NIMBLE study; the pozelimab monotherapy group was included in the study to evaluate the contribution of components only. †Defined as a ≥3-point reduction (improvement) in MG-ADL total score from baseline at week 24. ‡Defined as a ≥5-point reduction (improvement) in QMG total score from baseline at week 24. §Not statistically significant at prespecified final α=0.0459; the remaining α was used in an interim analysis. ¶p values not reported as secondary endpoints were subject to hierarchical testing and the previous endpoint was not statistically significant. ||Defined as an MG-ADL total score of 0 or 1 at week 24. **Defined as a ≥2-point reduction (improvement) in MG-ADL total score on two or more consecutive assessments spanning 4 or more weeks during the double-blind treatment period.

Table 2: Primary and secondary efficacy endpoints (modified intention-to-treat primary analysis set)

reason and the lowest total number of days of hospitalisation for any reason versus the other treatment groups. All participants who were hospitalised in the placebo group had at least one admission related to myasthenia gravis, while the cemdisiran, combination, and pozelimab groups each had one hospitalisation related to myasthenia gravis.

The least-squares mean differences in change from baseline in MG-ADL total score at week 24 were 0.2 (SE 0.6; 95% CI -1.0 to 1.4; $p=0.74$) for combination therapy versus cemdisiran monotherapy (assessed in the mITT analysis set) and -0.9 (0.7; -2.3 to -0.5; $p=0.22$) for combination therapy versus pozelimab monotherapy (assessed in the mITT-PC analysis set). Thus, combination therapy showed no added benefit over cemdisiran monotherapy.

Despite its clinical efficacy, cemdisiran monotherapy showed a lower percentage of terminal complement inhibition at week 24 than the combination therapy (figure 2G). Mean percentage change from baseline to week 24 in CH50 was -76.6% (SD 15.9) in the cemdisiran group, -99.6% (2.4) in the combination group, and 4.1% (25.7) in the placebo group. In participants who received cemdisiran monotherapy, a change of approximately -55.8% (15.2) from baseline in CH50 inhibition was observed by week 2 and reached a peak effect of around -76.8% (14.6) change from baseline by week 4, which was sustained throughout the double-blind treatment period. In participants who received combination therapy, an approximately -83.5 (18.0) change from baseline in inhibition of CH50 was observed by week 2 and progressed to around -98.9 (4.6) by



(Figure 2 continues on next page)

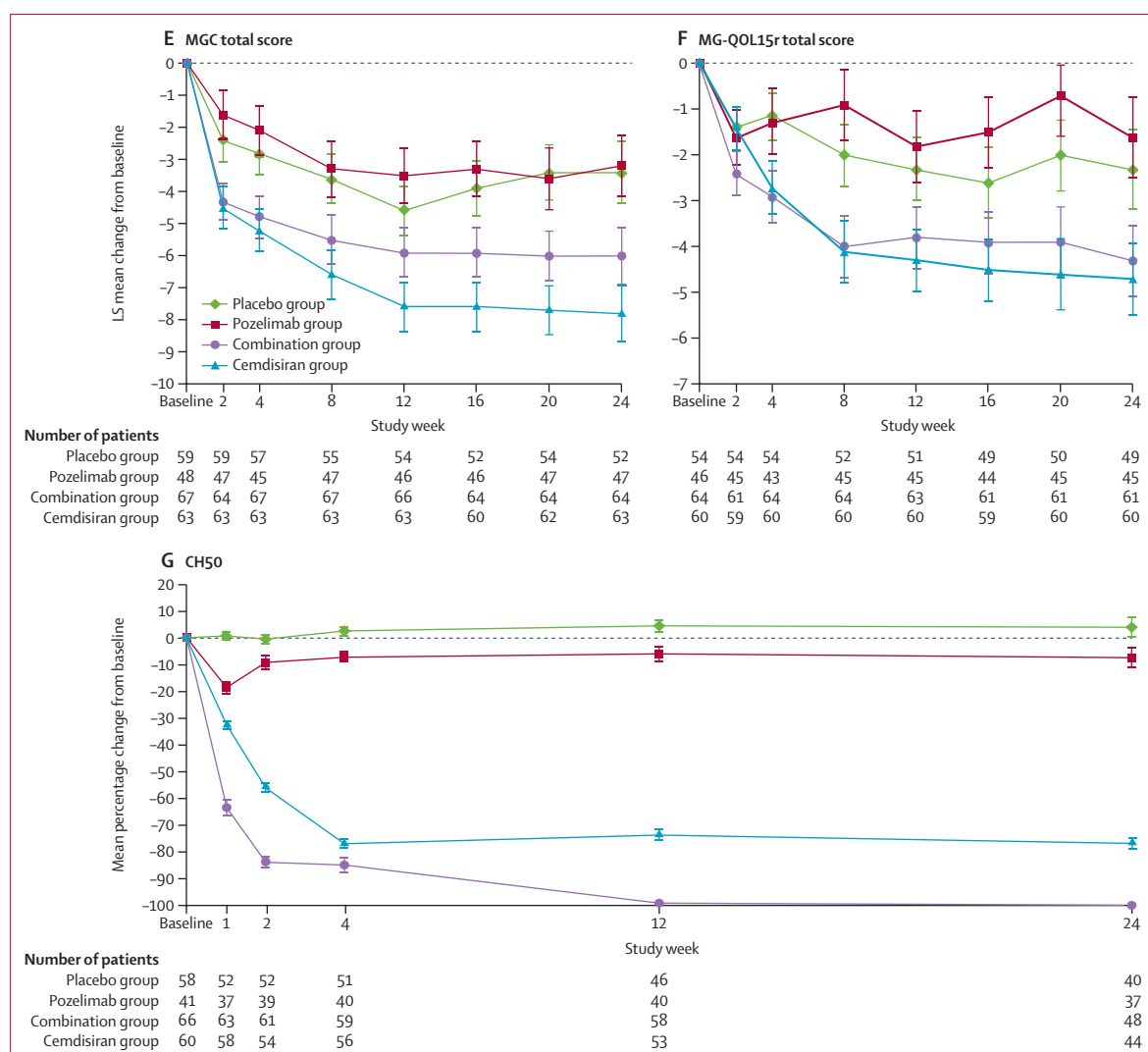


Figure 2: Patient-reported outcomes, performance measures, and CH50 during the double-blind treatment period (modified intention-to-treat primary analysis set)

(A, B) Change from baseline in MG-ADL total score (A) and QMG total score (B) over time; error bars are SEs. (C, D) Proportion of participants with improvement in MG-ADL total score (C) and QMG total score (D) from baseline by week 24. (E, F) Change from baseline in MGC total score (E) and MG-QOL15r total score (F) over time; error bars are SEs. (G) Percentage change from baseline in CH50 over time; error bars are 95% CIs. CH50=total complement haemolysis activity. LS=least-squares. MG-ADL=Myasthenia Gravis—Activities of Daily Living. MGC=Myasthenia Gravis Composite. MG-QOL15r=Myasthenia Gravis Quality of Life 15-item (revised). QMG=Quantitative Myasthenia Gravis.

week 12, which was sustained throughout the double-blind treatment period. The efficacy benefit was seemingly uncoupled from the degree of complement inhibition even at the earliest timepoints—ie, the efficacy of cemdisiran monotherapy matched or exceeded that of the combination at week 2 in most measures, despite showing a smaller change from baseline in CH50 inhibition (figure 2).

Total C5 concentrations are presented in appendix 1 (pp 19, 27).

The proportion of participants with at least one adverse event during the double-blind treatment period was 54 (69%) of 78 in the cemdisiran group,

65 (81%) of 80 in the combination group, 40 (82%) of 49 in the pozelimab group, and 54 (77%) of 70 in the placebo group (table 4). Most adverse events were mild to moderate in severity. The most common adverse event in the cemdisiran group was upper respiratory tract infection (nine [12%] of 78 patients), which occurred at a similar incidence in the placebo group (eight [11%] of 70). Overall, the rates of infection during the double-blind treatment period were 21 (27%) of 78 in the cemdisiran group, 30 (38%) of 80 in the combination group, and 28 (40%) of 70 in the placebo group. There were no meningococcal infections in any group during the double-blind treatment period.

	Placebo (n=66)	Combination (n=75)	Cemdisiran (n=75)	Pozelimab (n=49)
Participants with at least one myasthenic crisis (including impending crisis)	9/66 (14%)	6/75 (8%)	1/75 (1%)	0/49
Participants with at least one day of any hospitalisation post-randomisation*	10/66 (15%)	7/75 (9%)	2/75 (3%)	5/49 (10%)
Total days of hospitalisation post-randomisation	117	60	10	50
Participants administered rescue therapy†	6/59 (10%)	4/67 (6%)	1/64 (2%)	1/49 (2%)

Data are n/N (%) or n. *All ten participants in the placebo group had myasthenia gravis-related hospitalisations; one participant in each of the combination, cemdisiran, and pozelimab groups had myasthenia gravis-related hospitalisations. †The modified intention-to-treat primary analysis set is presented for this endpoint only as rescue therapy was considered as an intercurrent event in efficacy analyses.

Table 3: Exploratory endpoints during the double-blind treatment period (modified intention-to-treat analysis set)

	Placebo (n=70)	Combination (n=80)	Cemdisiran (n=78)	Pozelimab (n=49)
Any adverse event	54 (77%)	65 (81%)	54 (69%)	40 (82%)
Mild	26 (37%)	35 (44%)	31 (40%)	12 (24%)
Moderate	21 (30%)	22 (28%)	21 (27%)	23 (47%)
Severe	7 (10%)	8 (10%)	2 (3%)	5 (10%)
Severe adverse event	7 (10%)	8 (10%)	2 (3%)	5 (10%)
Serious adverse event	10 (14%)	7 (9%)	2 (3%)	5 (10%)
Adverse event of special interest*	2 (3%)	12 (15%)	8 (10%)	1 (2%)
Treatment-related adverse event	16 (23%)	32 (40%)	23 (29%)	13 (27%)
Adverse event leading to study treatment discontinuation	2 (3%)	0	0	1 (2%)
Adverse event leading to dose interruption or reduction	3 (4%)	3 (4%)	1 (1%)	1 (2%)
Adverse events of infections and infestations by SOC†				
Serious infections and infestations	3 (4%)	2 (3%)	0	2 (4%)
Adverse events occurring in ≥5% of participants in any treatment group, by preferred term				
Upper respiratory tract infection	8 (11%)	6 (8%)	9 (12%)	6 (12%)
Headache	7 (10%)	9 (11%)	4 (5%)	5 (10%)
Nasopharyngitis	3 (4%)	2 (3%)	4 (5%)	2 (4%)
Rash	1 (1%)	2 (3%)	4 (5%)	2 (4%)
Urinary tract infection	2 (3%)	5 (6%)	4 (5%)	4 (8%)
Injection-site reaction	1 (1%)	6 (8%)	3 (4%)	1 (2%)
Diarrhoea	5 (7%)	11 (14%)	2 (3%)	6 (12%)
Arthralgia	1 (1%)	5 (6%)	1 (1%)	1 (2%)
Cough	1 (1%)	4 (5%)	1 (1%)	1 (2%)
Myasthenia gravis	12 (17%)	4 (5%)	1 (1%)	4 (8%)
Pain in extremity	1 (1%)	4 (5%)	1 (1%)	0
Pruritus	0	4 (5%)	0	0

Data are n (%) of patients with at least one event. No adverse events leading to death were recorded during the double-blind treatment period. ALT=alanine aminotransferase. SOC=system order class. ULN=upper limit of normal. *Defined as any hypersensitivity reactions potentially related to study treatment, suspected *Neisseria* infection, any injection-site reactions, and any of the following abnormalities: ALT ≥3 × ULN, if baseline ALT was less than the ULN; ALT ≥2 × baseline, if baseline ALT was at or above the ULN; or total bilirubin ≥2 × ULN considered hepatic in origin by the investigator. †No cases of suspected *Neisseria* (meningococcal) infection were recorded.

Table 4: Adverse events during the double-blind treatment period (safety analysis set)

23 participants had an adverse event of special interest, among which the most common were hypersensitivity and injection-site reaction in the cemdisiran and combination groups. All hypersensitivity and injection-site reactions were non-serious and mild to moderate in intensity, and none led to treatment discontinuation (appendix 1 p 23). One participant who had a history of autoimmune hepatitis and primary biliary cholangitis at baseline in the cemdisiran group had a liver alanine aminotransferase elevation, which was not related to the study treatment.

The proportion of participants with at least one serious adverse event during the double-blind treatment period was two (3%) of 78 in the cemdisiran group, seven (9%) of 80 in the combination group, and ten (14%) of 70 in the placebo group (table 4). Three serious adverse events occurred in two participants in the cemdisiran group: uterine leiomyoma and adenomyosis (one participant); and decreased oxygen saturation (one participant). These events were severe in intensity and were considered not related to the study treatment. No serious infections occurred in the cemdisiran group during the double-blind treatment period. In the placebo group, most serious adverse events were related to myasthenia gravis crisis, impending crisis, or myasthenic exacerbations. Serious adverse events considered related to study treatment were pulmonary embolism and deep vein thrombosis in the combination group (one patient), cellulitis in the placebo group (one patient), and urinary tract infection in the placebo group (one patient).

No deaths occurred during the double-blind treatment period. After the double-blind treatment period, two deaths occurred. One death due to pneumonia—assessed as related to the study treatment by the investigator but not related to the study treatment by the sponsor—occurred in a participant in the cemdisiran group during the extension treatment period. The other death was due to septic shock—assessed as not related to the study treatment by both the investigator and the sponsor—and occurred in an older participant in the combination group during the open-label treatment period. Additional details on the two deaths are described in appendix 1 (p 19).

Transient treatment-emergent antidrug antibody low titre responses to pozelimab and cemdisiran were observed in one (2%) participant who received combination treatment and one (2%) who received cemdisiran.

Mean concentration of total C5 in plasma at week 24 is presented in appendix 1 (p 27). Mean concentrations of cemdisiran and its major active metabolite, AS(N-2)3' cemdisiran, in plasma were quantifiable within 1–4 h after administration and fell below the lower limit of quantification in subsequent samples at week 12 (appendix 1 pp 28–29). Concentration-time profiles of pozelimab are presented in appendix 1 (p 30).

Discussion

This phase 3 trial showed the efficacy and safety of cemdisiran (a novel GalNAc-conjugated siRNA targeting C5 mRNA) as a monotherapy administered subcutaneously every 3 months, as well as in combination with a low dose of pozelimab (a fully human antibody targeting the C5 protein) administered subcutaneously every month, for the treatment of generalised myasthenia gravis. The cemdisiran monotherapy and combination groups both met the trial's prespecified primary and key secondary endpoints of improvement in the patient-reported outcome of MG-ADL and the physician-reported outcome of QMG versus placebo by week 24. The proportions of patients with MG-ADL and QMG responses by week 24 were also significantly greater with cemdisiran monotherapy and combination therapy than with placebo. The clinical benefits were rapid, sustained, and clinically meaningful, with no waning of efficacy near the end of the respective dosing intervals. Exploratory analyses also indicated that occurrence of generalised myasthenia gravis crisis (including impending crisis), use of rescue therapy, and hospitalisations were lower with cemdisiran monotherapy and combination therapy than with placebo.

A unique aspect of this study design allowed assessment of the contribution of each treatment component, which revealed that combination therapy had no additional benefit over cemdisiran monotherapy. As expected, the low dose of pozelimab monotherapy in this trial had little efficacy. Cemdisiran monotherapy reached a clinically meaningful, least-squares mean placebo-adjusted difference of -2.3 points on the MG-ADL (the primary endpoint), which was greater than the difference seen in the combination group (-1.7). No formal test of cemdisiran monotherapy superiority over combination therapy was conducted in this study; however, cemdisiran monotherapy results were generally numerically favourable for most clinical endpoints at almost every timepoint assessed. These efficacy findings were surprising because cemdisiran monotherapy led to a lower proportion of terminal complement inhibition ($\sim 77\%$) than combination therapy ($\sim 99\%$), as measured by CH50 assay; these relative degrees of complement inhibition were consistent with previous cemdisiran clinical trials.^{16,35}

Cemdisiran monotherapy was generally well tolerated, with no adverse events leading to treatment discontinuation, and no new safety concerns were identified. After the double-blind treatment period, there was one possible treatment-related death, as assessed by the investigator, in a patient in the cemdisiran group with diabetes and use of immunosuppressive therapies, which should be noted; the potential risk of infections was monitored carefully in patients receiving this treatment. There were no serious infections in the cemdisiran monotherapy group and no meningococcal infections in any group. Overall, adverse events and

infections were numerically lowest with cemdisiran monotherapy. The finding that cemdisiran monotherapy was also associated with lower rates of infection than the combination group might be attributed to the finding that it only partially inhibited complement compared with the combination therapy.

The different degree of efficacy found for cemdisiran monotherapy in generalised myasthenia gravis compared with that previously seen for cemdisiran monotherapy in PNH^{16,23} might be the most surprising aspect of our findings. In PNH, cemdisiran monotherapy had little activity, whereas the combination therapy, tested head-to-head versus ravulizumab in a small exploratory cohort of an ongoing phase 3 study (by lowering the C5 load and then also adding a blocking antibody) seemed to have better efficacy than current C5 inhibitors.²⁴ These findings would suggest that PNH requires complete complement inhibition for optimal disease control (as achieved by the siRNA-antibody combination) while generalised myasthenia gravis might benefit from substantial but only partial complement inhibition (as achieved by the siRNA alone). This could be due to the differing ways in which complement is activated and mediates damage in these two diseases. In PNH, the erythropoietic lineage is increasingly overtaken by mutated clones missing endogenous complement regulators normally found on the cell membrane, leading to haemolysis; the clinical data suggest that complete complement inhibition might be required to prevent intravascular haemolysis.¹⁶ By contrast, in generalised myasthenia gravis, autoantibodies activate complement and cause damage at the neuromuscular junction in the setting of normal endogenous regulators; the clinical data presented herein suggest that substantial but incomplete complement blockade might be sufficient to prevent neuromuscular junction damage, performing at least as well as complete complement blockade.

The underlying mechanisms explaining the differing degrees of complement inhibition required for efficacy in generalised myasthenia gravis versus PNH remain to be elucidated. Peripheral CH50 might not fully reflect complement activity at the neuromuscular junction, where disease pathology actually occurs. We also measured total C5 (appendix 1 p 27) and modelled free C5 (pharmacokinetic and pharmacodynamic modelling; unpublished data), which showed free C5 reduction of $\sim 98\%$ with cemdisiran. Based on the mechanism of action, cemdisiran suppresses the expression and secretion of C5 but does not eliminate the circulating residual free C5. This residual C5 might still be activated, generating C5a (anaphylatoxin) and C5b. Because C5b is necessary for the assembly of the C5b-9 membrane attack complex, even a small amount of residual free C5 ($\sim 2\%$) could be sufficient to support approximately 24% residual complement activity in the CH50 assay. By contrast, the two components used in combination offer a dual mechanism of action: cemdisiran reduces C5

production, while pozelimab blocks the remaining circulating C5, thereby fully depleting free C5 available for complement pathway activation.

The ability to achieve compelling efficacy with cemdisiran monotherapy, while maintaining residual classical complement activity, could also help explain the favourable safety profile observed in this study for monotherapy as compared with combination therapy. If confirmed in further studies, this would have important implications for the risk–benefit ratio in patients—ie, if disease control can be achieved without completely blocking complement activity, then the residual complement activity could also allow for lower infection risk. This finding suggests there might be an optimal amount of residual complement activity for other autoantibody disorders characterised by aberrant complement activity, which should be explored on a per-disease basis.

A limitation of this trial is that it was not designed to evaluate switching from target therapies to cemdisiran monotherapy. However, as shown in an ongoing phase 3 study of PNH in which ravulizumab-treated patients were switched to cemdisiran-based therapies,²⁴ there is no safety issue expected upon switching therapies. Additionally, the current findings are constrained by insufficient long-term safety and efficacy data, which will be available through extended monitoring up to 120 weeks upon study completion. Another limitation is that the study was not powered and analysed to definitively assess the superiority of cemdisiran monotherapy over combination. Subgroup analyses were not presented due to the small sample size in some subgroups (eg, history of positivity for anti-LRP4 antibodies; Black or African American race), which warrants future investigations. Finally, the magnitude of the placebo effect observed in this study was large, but is consistent with that in other phase 3 studies in generalised myasthenia gravis.^{7–9}

In summary, this phase 3 trial showed that cemdisiran monotherapy led to rapid and sustained improvements versus placebo in the treatment of symptomatic generalised myasthenia gravis, with an acceptable safety profile in the majority of patients. Cemdisiran subcutaneous dosing every 3 months was efficacious while preserving residual complement activity, with no meningococcal infections and no evidence of increased infective risk compared with placebo. Cemdisiran thus has the potential to offer a novel therapeutic option in the treatment arsenal for generalised myasthenia gravis.

Contributors

TV, HM, JV, AM, RP, UC, MD, ASH, LP, GDY, and JFH were involved in the conceptualisation of the study and interpretation of data analysis. TV, RP, UC, NJ, MD, WM, KAM, MEB, SS, SK, DP, NS, and ASO accessed and verified the data. All authors had access to all the included data and were critically involved in the interpretation of the data. RP and UC wrote the original draft. All authors reviewed and edited the manuscript, reviewed the final version of the manuscript, and accept responsibility for the decision to submit the manuscript for publication.

Declaration of interests

TV has served as the University of South Florida Site Principal Investigator for myasthenia gravis clinical trials sponsored by Alexion/AstraZeneca Rare Disease, Amgen, argenx, Cartesian Therapeutics, COUR, Dianthus Therapeutics, EMD Serono, ImmunAbs, Immunovant, Johnson & Johnson, NMD Pharma, Novartis, Regeneron Pharmaceuticals, Sanofi, VOR, and UCB; has served as a speaker for Alexion/AstraZeneca Rare Disease, Amgen, argenx, CSL Behring, and Johnson & Johnson; has served on advisory boards and/or received consulting fees from Alexion/AstraZeneca Rare Disease, argenx, Amgen, Dianthus Therapeutics, Johnson & Johnson, ImmunAbs, NMD Pharma, Takeda, and VOR; and has participated in continuing medical education events sponsored by AcademicCME, CMEO, PER, and MedLive. AAH has received research grants from Alexion/AstraZeneca, argenx, UCB, Immunovant, Regeneron Pharmaceuticals, CabalettaBio, Amgen, Novartis, Arcellx, Merck, Mkarta, NMDPharma, Janssen/Johnson & Johnson, Cour Pharmaceuticals, and Kyverna; has received consulting fees from Alexion/AstraZeneca, argenx, UCB, Regeneron Pharmaceuticals, Amgen, Novartis, Merck, NMDPharma, Grifols, Janssen/Johnson & Johnson, and Kyverna; has received honoraria from Alexion/AstraZeneca, argenx, UCB, Amgen, Janssen/Johnson & Johnson; has received support for attending meetings and/or travel from Alexion/AstraZeneca, argenx, UCB, Amgen, Janssen/Johnson & Johnson; has participated in a data safety monitoring board or advisory board for Alexion/AstraZeneca, argenx, UCB, Regeneron Pharmaceuticals, Amgen, Novartis, Merck, NMDPharma, Grifols, Janssen/Johnson & Johnson, and Kyverna; and received equipment, materials, drugs, medical writing, gifts or other services from Alexion/AstraZeneca, argenx, and UCB. SJ has served as an advisory board member for Alexion, Alnylam, Argenx, Immunovant, Johnson & Johnson, Merck, Novartis, Regeneron Pharmaceuticals, and UCB; served as an expert panel for argenx; and is a trustee of Myaware. HM has received research grants from Japan Ministry of Health and Labour and Welfare; has received consulting fees from Alexion, argenx, and UCB; and has received honoraria for speaking from Novartis, Johnson & Johnson, and Merck. JV has received grants/contracts from Novo Nordisk foundation; has received consulting fees from Sanofi, Dyne Therapeutics, and Roche; has received honoraria from Edgewise Therapeutics, UCB, Alexion, and Janssen; and participated in a data safety monitoring board for Dyne Therapeutics. ASH is stockholder of Regeneron Pharmaceuticals. AM has received grants/contracts from German Research Foundation, Alexion, argenx, Octapharm, and UCB; has received consulting fees from Alexion, argenx, Janssen, Novartis, and UCB; has received honoraria from Alexion, argenx, Amgen, Axunio, Desitin, Genpharm, Janssen, Neopharm, Novartis, Sanofi, and UCB; has participated in a data safety monitoring board or advisory board for Alexion, argenx, Amgen, Janssen, and Merck; and has served as a Chairman of the Association for Research into Myasthenic Syndromes in Germany and a member of the advisory board for German Myasthenia Gravis Society. PS has received consulting fees from Biogen and Sanofi; has received honoraria from Alnylam, AstraZeneca, Biogen, Novartis, and Roche; and has participated in a data safety monitoring board or advisory board for Biogen and Sanofi. SL has received grants/contracts from National Key Research and Development Plan of the National Natural Science Foundation of China. RP, UC, NJ, MD, WM, KAM, MEB, SK, SS, DP, NS, ASO, ASi, and LP are employees and stockholders of Regeneron Pharmaceuticals. GDY is an employee and stockholder of, has issued and pending patents within, and serves a leadership role in Regeneron Pharmaceuticals. JFH has received consulting fees from Regeneron Pharmaceuticals. RM, TL, YH, AA-R, JJ, and AG declare no competing interests.

Data sharing

Qualified researchers can request access to study documents (including the clinical study report, study protocol with any amendments, blank case report form, and statistical analysis plan) that support the methods and findings reported in this Article. Individual anonymised participant data will be considered for sharing under the following conditions: once the product and indication have been approved by major health authorities (eg, FDA, EMA, PMDA, etc) or development of the product has been discontinued globally for all indications on or after April, 2020,

and there are no plans for future development; if there is legal authority to share the data; and if there is not a reasonable likelihood of participant re-identification. Requests should be submitted at <https://vivli.org/>.

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